The allegory axioms, and what defines them in the Frobenius calculus

Freyd, *Categories, Allegories* §2.11–§2.112 against `functorialSemanticsForRelationalTheories.pdf` Def. 4.1
Lean: `Freyd/S2_10.lean` (`Allegory`), `diag/CB.lean` (`CartBicat`), `diag/CB_Allegory.lean` (`allegoryOfCartBicat`)

Where the pictures come from. Every picture of a theorem is emitted by `./scripts/diag-export` from that theorem's Lean type. Nothing about a statement is redrawn here. The two figures marked *transcribed* are the exception: they copy the paper's definition pictures, and a definition has no statement to export.

Equation numbering. The numbers below are the 2017 preprint's. The `CartBicat` fields no longer encode them — a field is named for what it says, `«∇Δ≤𝟙»` rather than `ineq_37` — and each docstring carries the preprint number. The repo's copy of the paper is now the published version — Poly. J. Math. **2** (6), 20 Oct 2025, doi:10.69763/polyjmath.2.6 — which renumbers. Translation, published ⟵ preprint: (34)⟵(37), (35)⟵(38), (36)⟵(39), (37)⟵(40), (38)⟵(41), (39)⟵(42), (40)⟵(43), (33)⟵(35), and Example 2.3(c)'s (13),(14)⟵(11),(12); the preprint's (36) is unnumbered there. Pages: Def. 4.1 at 20–21 (was 17), the † notation at 21 (was 19), Lemma 4.2 at 21–22 (was 19). Renumbering the Lean fields is a separate decision and has not been made.

Freyd's allegory takes reciprocation `°` and intersection `∩` as **primitive operations** and lists eight **equational axioms** on them. A cartesian bicategory of relations takes neither: every object carries one special Frobenius structure `Δ`, `!`, `∇`, `?`, the hom-order `≤` is primitive, and both `°` and `∩` are **defined** from the four dots. Every one of Freyd's eight axioms is then a theorem. The table below is that dictionary, one row per axiom.

Reading the pictures: left to right is composition — the book's diagram order, `R S` means first `R` then `S` — vertical stacking is `⊗`, a bare wire is an identity, a white box is a relation, and a box that is mirrored and tinted is its converse — the mirroring is the paper's own notation, the tint is this note's, because at a glance a chamfer is easy to miss. The label inside is **not** mirrored: the paper can reverse a single letter and still be read, a whole term like `R ∪ S` cannot. All four dots are solid, as Def. 4.1 draws them. The exporter draws a meet or a converse as a **labelled** block, so a composite under a `°` appears as a single box reading `R ; S`; a `⊗` draws as two stacked runs; and any subterm with no shape in this calculus — an associator, say — degrades to an opaque box. The term columns write the converse `R°`, Freyd's symbol, where the paper writes `R†`: these terms are read against the book throughout, and one symbol per idea beats asking the reader to match two. The Lean sources write `R°` as well. The paper reserves `(–)°` for the colour swap of its §7, but §7 is a worked example in other models — see the caution at the end of this note — and nothing here formalises it, so in a repo that is `Rel(Set)` throughout there is no clash to protect against.

In a chain table the symbol at the head of each row is the relation **that step was proved at**: = for a reshaping, ≤ for a genuine inequality. Counting them is the point of drawing a proof at all — a chain of one ≤ between equalities says the theorem is one law and a lot of bookkeeping.

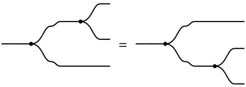
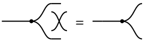
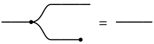
1. Definition 4.1, as the paper states it

`functorialSemanticsForRelationalTheories.pdf` Def. 4.1, pp. 20–21 — transcribed here as `class CartBicat` (`diag/CB.lean`), field for field. Every picture below is exported from the Lean field, so the class and the definition cannot drift apart.

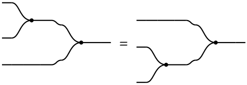
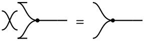
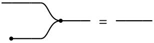
A **cartesian bicategory of relations** is a poset-enriched category, symmetric monoidal, satisfying four conditions. Clauses 1 and 2 are the two structures every object carries; clause 3 makes each the other's adjoint and imposes Frobenius; clause 4 is the only condition on arrows.

The tower this note describes is STRICT: the two bracketings of a threefold product are the same object, so no associator or unitor appears in any picture. The coherence arrows the paper suppresses in its own diagrams are absent here because they do not exist.

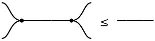
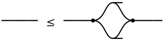
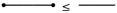
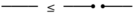
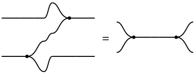
1.1. Every object carries a cocommutative comonoid `(Δ, !)`

	(8) **coassociativity** `CartBicat.Δ_assoc` — copying twice on the left is copying twice on the right.
	(9) **cocommutativity** `CartBicat.Δ_comm` — the two copies are interchangeable.
	(10) **counit** `CartBicat.Δ_counit` — copy, then discard one copy, and nothing has happened.

1.2. Every object carries a commutative monoid `(∇, ?)`

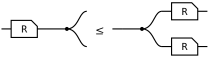
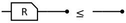
	(5) **associativity** `CartBicat.«∇_assoc»`.
	(6) **commutativity** `CartBicat.«∇_comm»`.
	(7) **unit** `CartBicat.«∇_unit»` — create, then merge, and nothing has happened.

1.3. Five inequations relate the monoid and the comonoid

	(37) `∇ ; Δ ≤ 𝟙` — half of `Δ ⊣ ∇`
	(38) `𝟙 ≤ Δ ; ∇` — the other half
	(39) `? ; ! ≤ 𝟙` — half of `! ⊣ ?`
	(40) `𝟙 ≤ ! ; ?` — the other half
	(41) the **Frobenius** law `CartBicat.frob_left`, and its mirror `frob_right`. This is the one equation of the group; the other four are inequalities.

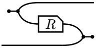
The **special** law `Δ ; ∇ = 𝟙` is NOT part of the definition. The paper derives it — “one direction is given by (38) and the other is proved as follows” — and so does the Lean, as `Freyd.Diag.CartBicat.special`; its picture is in the correspondence table below.

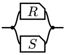
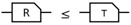
1.4. Every arrow is a lax comonoid homomorphism

	(42) `R ; Δ ≤ Δ ; (R ⊗ R)` `CartBicat.lax_Δ` — running `R` once and copying is below copying and running it on both strands.
	(43) `R ; ! ≤ !` `CartBicat.lax_!`.

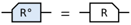
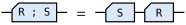
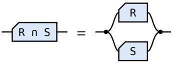
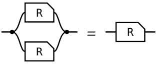
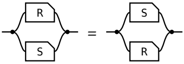
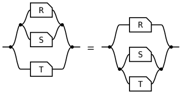
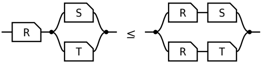
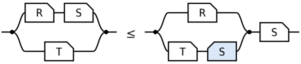
Read as inequalities in one direction only: an arrow that satisfied these as EQUATIONS would be a map, and the whole point is that a general relation is not.

2. Freyd’s primitives are the Frobenius calculus’s definitions

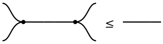
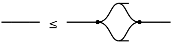
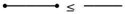
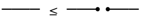
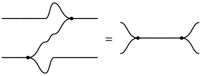
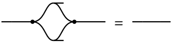
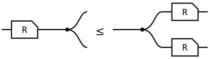
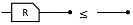
Freyd’s primitive, and what defines it here	**picture**
reciprocation `R°` — §2.11, primitive. Here `conv R := bend ((R ⊗ 𝟙) ; cap)` (`diag/CB.lean:223`), not a generator. Input arrives bottom left and runs under `R`; the **cup** `? ; Δ` at the top left creates a pair, one strand feeding `R` and the other leaving as the output at the top right; the **cap** `∇ ; !` at the bottom right annihilates `R`’s output against the input. It is the cup and the cap that reverse the arrow — a wire merely bumped up over the box and back down is an isotopy of `R` itself.	 transcribed

intersection `R ∩ S` — §2.11, primitive. Here the convolution `meet R S := Δ ; (R ⊗ S) ; ∇` (`diag/CB_Derived.lean:24`). Copy the input, run both, merge; the merge forces the two results to agree.	 transcribed
containment `R ⊑ S` — §2.11, **defined** as `R ∩ S = R`. Here the poset-enrichment ≤ is primitive and ∩ is derived, so the two directions are swapped. `meet_eq_left_of_le` (`diag/CB_Derived.lean:196`) is the bridge.	no picture — this is the 2-cell itself
maximal morphism `⊤` — §2.15, and Freyd needs a **unitary** allegory for it. Here `top a b := ! ; ?` (`diag/CB_Derived.lean:28`) is free in Def. 4.1. `Freyd.Diag.«≤_top»`.	

3. The eight axioms, and what proves each

Freyd's axiom, and what supplies it	picture, from the Lean statement
`(R°)° = R` — §2.11 `recip_recip`; here the **snake**, from (41). `Freyd.Diag.CartBicat.conv_conv`. Bending down and back up straightens out, so bending twice is the identity.	
`(R S)° = S° R°` — §2.11 `recip_comp`; here mirroring the picture. `Freyd.Diag.CartBicat.conv_comp`. Reflecting a chain left to right reverses it; each box comes back flipped.	
`(R ∩ S)° = R° ∩ S°` — §2.11 `recip_inter`; the same mirroring. `Freyd.Diag.conv_inter`. Δ and ∇ are each other's mirror image, so reflecting the convolution leaves its shape alone and only turns the boxes round.	
`R ∩ R = R` — §2.11 `inter_idem`; here (42) and the **special** law. `Freyd.Diag.meet_idem`. Copy, run R twice, merge — (42) pushes R back through the copy and `Δ ; ∇ = 𝟙` closes the bubble.	
`R ∩ S = S ∩ R` — §2.11 `inter_comm`; here cocommutativity (9) and commutativity (6). `Freyd.Diag.meet_comm`. The two strands of Δ are interchangeable, and so are those of ∇.	
`R ∩ (S ∩ T) = (R ∩ S) ∩ T` — §2.11 `inter_assoc`; here coassociativity (8) and associativity (5). `Freyd.Diag.meet_assoc`. Either bracketing builds the same three-way copy tree, and dually the same merge tree.	
semi-distributivity `R (S ∩ T) ⊑ R S ∩ R T` — §2.11 `semidistrib`, §2.136; here (42), the **lax comonoid** inequation. `Freyd.Diag.semidistrib_of_lax`. Running R and then copying is below copying and then running R on each strand.	
modular identity `R S ∩ T ⊑ (R ∩ T S°) S` — §2.112 `modular`, the axiom Freyd **adjoins**: “we adjoin one more containment, the law of modularity”. Here (41) and (42), **derived**. `Freyd.Diag.modular_of_frobenius`. Strip the shared prefix `Δ ; (R ⊗ T)` off both sides and what is left is `«∇_slide_conv»` (`diag/CB.lean:560`) — a box slides through a merge at the cost of its converse on the other strand.	

4. The other direction: what each law of Def. 4.1 says in Freyd's language

picture, from the Lean statement	Def. 4.1	Freyd
	(37) `∇ ; Δ ≤ 𝟙` half of `Δ ⊣ ∇`	`Δ° Δ ⊑ 1` — **Δ is simple** §2.13. Not used anywhere in the bridge; see below.
	(38) `𝟙 ≤ Δ ; ∇` the other half of `Δ ⊣ ∇`	`1 ⊑ Δ Δ°` — **Δ is entire** §2.13; with (37), Δ is a **map**.
	(39) `? ; ! ≤ 𝟙` half of `! ⊣ ?`	`!° ! ⊑ 1` — **! is simple** §2.13. Also unused in the bridge.
	(40) `𝟙 ≤ ! ; ?` the other half of `! ⊣ ?`	`1 ⊑ ! !°` — **! is entire** §2.13; with (39), `!` is a map. This is what makes `⊤ = ! ; ?` contain every arrow.
	(41) the **Frobenius** law an axiom. `𝟙 ⊗ Δ` and `∇ ⊗ 𝟙` draw as stacked runs; only the associator `α⁻¹` stays an opaque box, having no shape in this calculus.	the **modular identity** §2.112 — and reciprocation §2.11 itself, since the cup and cap only satisfy the snake because of (41).
	the **special** law **not** an axiom: derived from (38) and (40), `Freyd.Diag.CartBicat.special`	`Δ Δ° = 1` — **Δ is injective**, so by §2.141 at `f = g = Δ` it is a monic map. In `Rel(Set)`: copying is total and loses nothing.
	(42) every arrow is a **lax comonoid homomorphism**	§2.136′s `R (S ∩ T) ⊑ R S ∩ R T`, read at `S := π₁°`, `T := π₂°` — Freyd's semi-distributivity, product-free.
	(43) every arrow is **lax for the counit**	§2.152′s step "`p_α` is maximal in `(α, λ)`, hence `R p_β ⊆ p_α`" — that is, `R ⊑ ⊤`.

5. Unitary and pre-tabular — the two conditions the converse needs

Freyd's §2.11 axioms are all a cartesian bicategory has to give back. Going the other way needs two further conditions, and they are exactly the two the repo has not formalised. Both definitions below are quoted from `Freyd/S2_10.lean`, which follows the book.

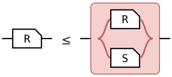
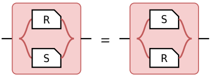
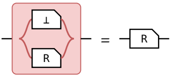
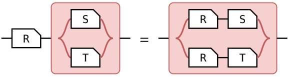
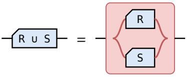
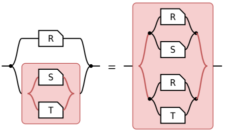
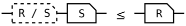
condition	Freyd	what the Frobenius side already has
unitary §2.15	`T` is a **partial unit** when `𝟙 T` is the largest endomorphism on it, `PartialUnit T := ∀ R : T → T, R ⊑ 𝟙 T` (`:539`); it is a **unit** when moreover every object carries an entire arrow to it, `IsUnit T := PartialUnit T ∧ ∀ a, ∃ R : a → T, Entire R` (`:545`). A `UnitaryAllegory` is an allegory with one (`:548`). The point of a unit is that it makes `T_{a,b}` exist: `T = p_α p_β°` through the unit, which is where §2.152′s "`R p_β ⊆ p_α`" — the repo's (43) — comes from.	**already there, for free.** The monoidal unit `𝕀` is the unit object, `!_a : a → 𝕀` is the entire arrow every object carries to it — (40) says so — and `⊤ = ! ; ?` is maximal by `Freyd.Diag.«≤_top»`. So every cartesian bicategory of relations is unitary, with nothing assumed. Not recorded in Lean: `allegoryOfCartBicat` builds a plain `Allegory`, and no `UnitaryAllegory` instance is derived from `CartBicat`.
pre-tabular §2.165	`f, g` **tabulate** `R` when both are maps, `R = f° g`, and `f f° ∩ g g° = 𝟙` (`Tabulates`, `:502`); `R` is **tabular** when some pair tabulates it (`:506`). An allegory is `PreTabularAllegory` when every morphism is contained in a tabular one (`:556`). §2.165: "If `A` is unitary then pre-tabularity is equivalent to the existence of tabulations for maximal morphisms."	**this is what ⊗ is.** Given unitarity, the condition reduces to tabulating each `⊤_{a,b}`, and a tabulation of `⊤_{a,b}` is precisely an object `a ⊗ b` with its two projections — the monoidal product Def. 4.1 assumes. That is the whole content of the swap: Freyd derives the product from tabulations, Def. 4.1 posits it. Not built. Rebuilding ⊗ from tabulations and products in `Map(𝒜)` is `cartBicatOfUnitaryPretabular`; `diag/PLAN.md` holds it back for want of a consumer.

Together they are the whole gap: **cartesian bicategory of relations ≃ unitary pre-tabular allegory** (`DiagrammaticAlgebraOfFirstOrderLogic` §10, after Carboni–Walters). One half of that equivalence is `allegoryOfCartBicat` and is a theorem in this repo; the other half is the two rows above.

6. Union and residual — what phases 8 and 9 add

`∪` and `/` are past where Freyd's §2.11 list stops, and past what Def. 4.1 can build: they need a **biproduct** on top of the Frobenius structure (`diag/Tape.lean`, after `TapeDiagrams.pdf`) and a **second composition** with linear adjoints (`diag/FO.lean`, after `DiagrammaticAlgebraOfFirstOrderLogic.pdf`). Both are built, so both draw.

The two notations say which layer a picture is in. A union is a **tape**: the rounded wrapper is the second monoidal product, and its ▹/◃ open and close a branch a particle takes exactly one of — the same fork-runs-join shape as a meet, on a different product. A residual is a **dashed** box, because it is not a composite of the generators at all: `R / S := R ⨾• S⊥` lives in the second composition, and a solid box would claim it was drawable in the first.

statement	**picture, from the Lean statement**
`R ⊑ R ∪ S` — §2.2 in Freyd; here `Freyd.Diag.Biprod.«≤_union_left»`. `union R S := ⟨R, S⟩ ; [𝟙, 𝟙]` (`diag/Tape.lean:84`): offer both branches, then forget which was taken.	
`R ∪ S = S ∪ R` — `Freyd.Diag.Biprod.union_comm`. The two branches of a tape are interchangeable, exactly as the two strands of Δ are.	
`⊥ ∪ R = R` — `Freyd.Diag.Biprod.bot_union`. `bot a b := toZero a ; (toZero b)°` (`diag/Tape.lean:87`), routed through the zero object; the exporter has no shape for it, so it prints as a box.	
`R (S ∪ T) = R S ∪ R T` — `Freyd.Diag.Biprod.comp_union`; composition distributes over union, which `∩` does not. Freyd §2.2′s distributivity, the reason a distributive allegory is more than an allegory.	
`(R ∪ S)° = R° ∪ S°` — `Freyd.Diag.Biprod.conv_union`. The converse is a 2-functor for the tape layer too.	
`R ∩ (S ∪ T) = (R ∩ S) ∪ (R ∩ T)` — `Freyd.Diag.FbCbRig.meet_union_distrib`, the law that makes `distributiveAllegoryOfFbCb` a **distributive** allegory (§2.2). The one picture with both layers in it: meets inside a tape on the left, tapes around meets on the right.	
`(R / S) S ⊑ R` — `Freyd.Diag.ClosedLinearBicat.«residual_comp_≤»`; the residual is a lower bound of the arrows it is the greatest of. Freyd's §2.31 division: `R / S` is the largest `T` with `T S ⊑ R`, and `le_residual_iff` (`diag/FO.lean:170`) is that Galois statement in full. The dashed frame marks the box as belonging to the second composition.	

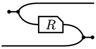
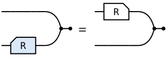
7. Lemma 4.2 (ii) — the proof, not just the statement

`diag-export --proof` draws a **proof**. A `calc` chain elaborates to nested `Trans.trans`, so its intermediate terms are all recoverable from the proof term: flatten the spine, infer the type of each leaf, draw one canvas per step. Nothing is replayed and nothing is retold — the nine rows below are the chain the Lean kernel accepted for `Freyd.Diag.CartBicat.conv_comp`, which is Lemma 4.2 (ii), `(R ; S)° = S° R°`.

7.1. First: why these pictures are not pictures of a converse

They are not, and that is the thing to get straight before reading them. The converse is drawn with **two** bends — a cup on the way in and a cap on the way out, each a pair of dots, `? ; Δ` and `∇ ; !` — which is the left-hand figure below. The proof never draws that. `conv_unique` says an arrow is **the** converse of R as soon as it satisfies one equation between UNBENT arrows `a ⊗ b → I`: two strands in on the left, one cap on the right. Bending is a bijection (`bend_unbend`/`unbend_bend`), so settling the unbent equation settles the bent one, and the cup never has to be drawn at all.

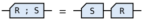
That is why every row below has a cap and no cup, and why an arrow that “is a converse” shows up not as a bent box but as a box sitting on the **bottom** strand, mirrored.

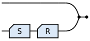
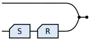
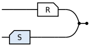
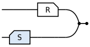
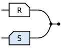
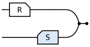
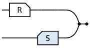
the converse, as the paper draws it	**the one rule the proof uses, twice**
	
R° is R with both wires bent round: in at the bottom left, cup at the top left, cap at the bottom right. Two dots at each bend, as `functorialSemanticsForRelationalTheories.pdf` draws it — a cup is ? then Δ, a cap is ∇ then !. **Transcribed by hand;** a definition has no statement to export.	`conv_slide` (`diag/CB.lean:449`): a box on the BOTTOM strand facing a cap is the same as that box on the TOP strand, upright. Sliding round the bend is what turns R° into R. This is the entire content of Lemma 4.2 (ii); the other seven steps are bookkeeping.

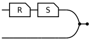
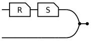
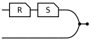
7.2. The chain

One row per **term**, not per step: the right-hand side of every step is the left-hand side of the next, so drawing steps would draw every picture twice. The first column is the term as the Lean has it, fully bracketed, with ⊗ binding tighter than `;` — it comes off the same walk as the picture, so the two cannot drift apart. Watch which **strand** each box is on: bottom means mirrored, top means upright.

Lemma 4.2 (ii), `(R ; S)° = S° ; R°`



term	**picture**	**the rule that reaches it**
`1 ⊗ (S° ; R°) ; cap`		**the start.** What `conv_unique` asks about S° R°: both boxes on the bottom strand, mirrored — that is what “these are converses” looks like once the wires are unbent.
`= 1 ⊗ S° ; (1 ⊗ R° ; cap)`	= 	**1.** ⊗ is a functor, so doing two things to one strand in sequence is the sequence of doing them. **Same picture.**
`= 1 ⊗ S° ; (R ⊗ 1 ; cap)`	= 	**2. `conv_slide`.** R° is the box nearest the cap, so it slides round the bend: it leaves the bottom strand, arrives on the top, and comes back upright as R. **The first of the two steps that prove anything.**
`= (1 ⊗ S° ; R ⊗ 1) ; cap`	= 	**3. `simp only [Cat.assoc]`.** Re-bracketing — visible in the term column, invisible in the picture. **Same picture.**
`= R ⊗ S° ; cap`	= 	**4. `tensHom_split'`** — interchange. R is on the top strand and S° on the bottom, so running them in sequence and running them side by side are the same thing. Shows as a re-spacing only because the exporter lays a diagram out left to right.
`= (R ⊗ 1 ; 1 ⊗ S°) ; cap`	= 	**5. `tensHom_split`** — the same law, back the other way. Split them apart again, R first, so that S° is now the box facing the cap. Which is what the next real step needs.
`= R ⊗ 1 ; (1 ⊗ S° ; cap)`	= 	**6. `simp only [Cat.assoc]`.** Re-bracketing. **Same picture.**

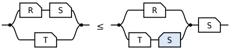
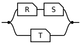
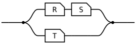
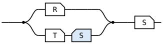
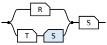
`= R ⊗ 𝟙 ; (S ⊗ 𝟙 ; cap)`	= 	**7. `conv_slide`,** on `S` this time. `S°` slides round the bend and arrives upright on the top strand. Both boxes are now on top, both upright, in the order `R` then `S`. **The second and last real step.**
`= (R ⊗ 𝟙 ; S ⊗ 𝟙) ; cap`	= 	**8. `simp only [Cat.assoc]`.** Re-bracketing. **Same picture.**
`= (R ; S) ⊗ 𝟙 ; cap`	= 	**9. `tensHom_comp`** — interchange once more; the two top-strand boxes merge into one. This is `R ; S` unbent, so `conv_unique` reads off `S° R° = (R ; S)°`. **Done.**

What the chain measures. Read down the pictures and most consecutive pairs are the same drawing — steps 1, 3, 6, 8 and 9 change nothing at all, and step 4 changes only spacing — while the term column beside them changes at every row. Those are associativity, the unit laws and interchange — precisely the laws a string diagram has built into its geometry rather than as rewrites. Seven of these nine Lean steps are bureaucracy the picture language does not charge for, and the theorem is two applications of `conv_slide`.

How this differs from the paper's proof. The paper proves (ii) by inserting a snake in the middle of `R ; S` and reading the two halves off as `S°` and `R°` — three pictures, all bent. The Lean never bends. Same theorem, different route, and the export shows the route the kernel accepted. Drawn by hand it would have shown the paper's three pictures and quietly implied those were what the repo had checked.

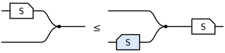
8. The modular law — proved, drawn

Freyd **adjoins** the modular identity: an allegory has no structure to derive it from, so it goes in as an axiom. Here it is a theorem, and this is its proof — `Freyd.Diag.modular_of_frobenius` (`diag/CB_Allegory.lean:48`), four terms, of which exactly one step is an inequality.

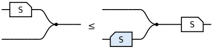
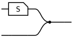
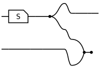
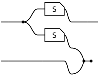
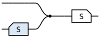
the modular law, `R S ∩ T ≤ (R ∩ T S°) S` 		
term	**picture**	**the rule that reaches it**
`(R ; S) ∩ T`		**the start:** `R S ∩ T`. A meet, so a copy and a merge with `R S` above and `T` below.
`= (Δ ; R ⊗ T) ; (S ⊗ 𝟙 ; ∇)`	= 	**Factor out the shared prefix.** `(R S) ⊗ T` is `(R ⊗ T) ; (S ⊗ 𝟙)`, so everything before the `S` is common to both sides and can be set aside. An equality, and drawn as one.
`≤ (Δ ; R ⊗ T) ; (𝟙 ⊗ S° ; (∇ ; S))`	≤ 	**«`∇_slide_conv`»,** under the shared prefix. `S` slides through the merge and reappears as `S°` on the other strand. **The only inequality in the proof** — everything else is reshaping.
`= R ∩ (T ; S°) ; S`	= 	**Fold the prefix back in.** `R ⊗ (T S°)` is `(R ⊗ T) ; (𝟙 ⊗ S°)`, and the copy-merge pair reads as a meet again. `(R ∩ T S°) S`, which is the goal.

8.1. Inside the middle step

«`∇_slide_conv`» (`diag/CB.lean:560`) is where the mathematics is, and it mentions only `S` — `R` and `T` sit untouched in the prefix throughout. Read on its own it says one arrow slides through a merge:



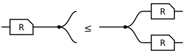
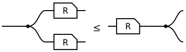
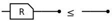
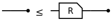
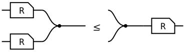
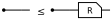
`S` occurs once on the left and twice on the right. That is the tell: the lax copy inequation (42) is the only law in Def. 4.1 that may duplicate a box, so it has to be the inequality step — and its own chain shows it is the only one.

«`∇_slide_conv`», `(S ⊗ 𝟙) ; ∇ ≤ (𝟙 ⊗ S°) ; ∇ ; S`		
term	**picture**	**the rule that reaches it**
`S ⊗ 𝟙 ; ∇`		**the start:** `(S ⊗ 𝟙) ; ∇.`
`= (S ; Δ) ⊗ 𝟙 ; 𝟙 ⊗ cap`	= 	**«`∇_of_cap`»,** backwards. The merge is rebuilt from a copy and a cap, which puts the term into the shape `(S ; Δ) ⊗ 𝟙` that (42) applies to.
`≤ (Δ ; S ⊗ S) ⊗ 𝟙 ; 𝟙 ⊗ cap`	≤ 	**(42), `lax_Δ`.** `S ; Δ` becomes `Δ ; (S ⊗ S)`: the box is copied. **The whole of the modular law, in one step.**
`= 𝟙 ⊗ S° ; (∇ ; S)`	= 	**«`cap_tens_∇`» and `conv_slide`.** Reshaping back, and the surviving duplicate slides round the cap to become `S°` — the same rule Lemma 4.2 (ii) is made of.

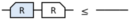
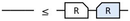
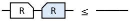
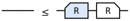
What the associators cost. α and ρ have no shape in this calculus and print as opaque boxes, so the two middle terms of the inner chain are less legible than its ends. That is the honest price of drawing what the Lean says: the reshaping steps really are that fiddly, and a hand-drawn version would have hidden them by working up to coherence.

9. Maps: SV, TOT, INJ, SUR, and the two inequations that always hold

Def. 4.1 clause 4 says every arrow is lax for `Δ` and for `!` — inequations (42) and (43). Applying Lemma 4.2 (ii) and (iv) to them gives (44) and (45), the same statements about `∇` and `?`. Those four hold for **every** arrow. Their four REVERSES do not, and each one failing or holding is a property of the arrow: that is (SV), (TOT), (INJ), (SUR).

	holds for every arrow	**holds iff the arrow is ...**
copy `Δ`	(42) `R ; Δ ≤ Δ ; (R ⊗ R)` 	(SV) — **single valued** 
discard `!`	(43) `R ; ! ≤ !` 	(TOT) — **total** 
merge `∇`	(44) `∇ ; R ≤ (R ⊗ R) ; ∇` — the converse of (42). **Not in the Lean;** see below.	(INJ) — **injective** 
unit `?`	(45) `? ; R ≤ ?` — the converse of (43). **Not in the Lean;** see below.	(SUR) — **surjective** 

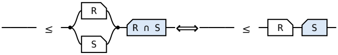
The paper gives the same four properties a second time, as ADJOINT conditions — (46)–(49) — and Lemma 4.4 says the two presentations agree. `diag/CB_Derived.lean` takes the adjoint forms as the definitions, because they are literally Freyd's `Simple` and `Entire` (§2.13), so a `diag` map and a book map are one predicate with no translation layer.

property	adjoint form, the definition	Freyd
single valued `Freyd.Diag.SingleValued`		`Simple R := R° R ⊑ 1` §2.13
total `Freyd.Diag.Total`		`Entire R := 1 ⊑ R R°` §2.13. With **single valued**, R is a **map**.
injective `Freyd.Diag.Injective`		`R R° ⊑ 1` §2.141 at `f = g = R`: a monic pair, so a monic map.
surjective `Freyd.Diag.Surjective`		`1 ⊑ R° R` §2.147: “`g` is a cover iff `1 ⊑ g° g`”, that is, iff `g°` is entire.

Why (44) and (45) have no picture. Both are converse-images, so drawing them from a Lean statement needs that statement to exist, and it does not. Getting it needs `Δ° = ∇`, `!° = ?` and Lemma 4.2 (iii) `(f ⊗ g)° = f° ⊗ g°` — and the first of those is a Frobenius computation at a COMPOSITE object, which is the clause of Carboni & Walters’ definition that Def. 4.1 as printed omits (`diag/CB.lean`’s header records this). It is the same gap that leaves Lemma 4.4 unproved in the repo. The four reverse inequations opposite are unaffected: they are recorded under the paper’s own labels in `diag/CB_Derived.lean`, and nothing downstream depends on Lemma 4.4.

10. Entireness of an intersection

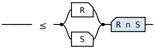
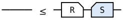
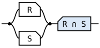
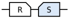
Two definitions from Freyd, both used below. The **domain** of `R` is the part of the identity where `R` is defined, `Dom R = 1 ∩ R R°` (§2.122). `R` is **entire** if it is defined everywhere, `Dom R = 1`, equivalently `1 ⊑ R R°` (§2.13) — which is `Total` above, the picture opposite (47). The question this section answers is when an **intersection** is entire: Freyd’s §2.124, “a lemma we will use repeatedly”, and B&dM 4.18.

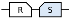


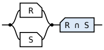
`Freyd.Diag.entire_inter_iff` (`diag/S2_124.lean:213`) — `Total (R ∩ S) ↔ 𝟙 ≤ R ; S°`

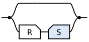
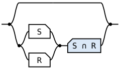
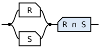
Read the two sides: on the left `R ∩ S` is named **twice**, on the right `R` and `S` are named **once each** and the meet is gone. That is the whole point of the lemma — the meet is needed to **ask** whether something is entire, not to **answer** it — and it is why every later use quotes the right-hand form. Freyd spends it at §2.147, where `H = x f° ∩ y g°` has to be shown entire; B&dM spend it throughout chapter 4.

`Freyd.Alg.entire_inter_iff` (`AOP/A4_2.lean:187`) is the same statement for a bare allegory, and there it is `Iff.rfl` on top of `dom_inter`: in `Freyd/S2_10.lean` the order `X ⊑ Y` is **defined** as `X ∩ Y = X`, so “`Dom(R ∩ S) = 1`” and “`1 ⊑ R S°`” are the same proposition once §2.124 is known. Here ≤ is primitive, so both directions are real steps — and they cost very different things.

⟹ given `R ∩ S` entire,  show 		
term	**picture**	**the rule that reaches it**
`𝟙`	——	**the start:** `𝟙`.
`≤ R ∩ S ; (R ∩ S)°`	≤ 	**the hypothesis:** `R ∩ S` entire. By the definition of entire, `𝟙 ≤ P P°` at `P = R ∩ S`.
`≤ R ; S°`	≤ 	**monotonicity, twice:** `R ∩ S ≤ R`, and `R ∩ S ≤ S` under `°`. No modular law.

⟸ given ——— ≤ 

show R ∩ S entire, ——— ≤ 

term	picture	the rule that reaches it
𝟙	———	**the start:** 𝟙.
≤ 𝟙 ∩ (R ; S°)	≤ 	**the hypothesis,** met with 𝟙. `meet_glb`: 𝟙 ≤ 𝟙 and 𝟙 ≤ R S°, so 𝟙 is below the meet of the two.
= 𝟙 ∩ (S ∩ R ; (S ∩ R)°)	= 	**§2.124:** 𝟙 ∩ R S° = 𝟙 ∩ (S ∩ R)(S ∩ R)°. R S° relates a to a′ when some b has a R b and a′ S b. The 𝟙 ∩ sets a′ = a, and it then reads “some b has a R b and a S b” — one arrow of R ∩ S. Two arrows become one; that is the step, and it is what needs the modular law. Freyd writes the right-hand side Dom (S ∩ R): Dom X is his name for 𝟙 ∩ X X°, so it is the same term, not an extra operation.
≤ R ∩ S ; (R ∩ S)°	≤ 	**X ∩ Y ≤ Y,** then S ∩ R = R ∩ S. The outer fork drops away, leaving the goal.

What the asymmetry says. One direction is two applications of monotonicity; the other needs §2.124, which needs the modular law, which in this calculus needs the Frobenius equation (41) and the lax copy inequation (42) — §7 above. So the picture chain locates the cost exactly: the third row of the second table is the whole strength of the lemma, and everything else is bookkeeping.

11. Anything missing?

Nothing on Freyd’s side. All eight §2.11 axioms are theorems of Def. 4.1, and `allegoryOfCartBicat` (`diag/CB_Allegory.lean:85`) discharges the `Allegory` structure field for field. The modular identity, which Freyd has to **adjoin** because an allegory has no structure to derive it from, comes out of (41) and (42).

Three things are stated but unproved. «∇Δ≤𝟙» and «?!≤𝟙» are discharged and never used (see below); (44) and (45) are not stated at all; and Lemma 4.4, that the comonoid and adjoint forms of the map conditions agree, is not proved. All three trace to one omission in Def. 4.1 as printed — no Frobenius structure at a composite object — which `diag/CB.lean`’s header records.

Two laws on the Frobenius side do no work here. Grepping the repo, «∇Δ≤𝟙» and «?!≤𝟙» are never used: they appear only in the class definition and in `diag/RelSetCB.lean`, where the `Rel(Set)` instance discharges them. They are there to make Δ ⊣ ∇ and ! ⊣ ? genuine adjunctions — which is what pins ∇ = Δ° and ? = !° by uniqueness of adjoints — not to prove any allegory law.

The bridge runs one way only, for the reason the section above sets out: a cartesian bicategory carries a symmetric monoidal ⊗ and an allegory carries nothing of the kind.

What Def. 4.1 alone does not have. Everything past Freyd’s §2.11 needs structure beyond it, and — this is the point — from **different sources**, not from `functorialSemanticsForRelationalTheories.pdf`:

construct	Freyd	diagrammatic counterpart	status
tabulation	§2.14	none	no counterpart. Def. 4.1 gives no splitting of coreflexives; ⊤ = ! ; ? is all the tabular content there is.
∪, ⊥	§2.2, distributive allegory	a **biproduct** ⊕ on top of Def. 4.1 — `Biprod`, `union`, `bot`, and `distributiveAllegoryOfFbCb` in `diag/Tape.lean`; `TapeDiagrams.pdf`	built, and drawn above. Not derivable from Def. 4.1 alone — the paper’s calculus has no union at all.
R / S, complement	§2.31, division allegory	`LinearBicat`, `ClosedLinearBicat`, `residual` — `diag/FO.lean`; `DiagrammaticAlgebraOfFirstOrderLogic.pdf`	built, and drawn above. The `Rel(Set)` instance (`diag/RelSetFO.lean`) needs `Classical.choice`.
Λ, power transpose	§2.4	none	open. No calculus in this tower has a power object.
Allegory ⟹ CB	§2.15 + §2.165	`cartBicatOfUnitaryPretabular`	deliberately not built.

One caution on §7–§9 of the paper, which look like they extend this table and do not: they are worked examples in **other** models. The paper says so itself at the end of §7 — only the black fragment `⊤, ∩, (−)†, ; , id` coincides with the calculus of relations there, while "`⊥` is not the empty relation and `>` is not the union".

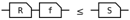
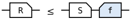
12. The shunting rule

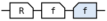
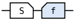
A map may be moved from one side of a containment to the other, at the cost of its converse. That is the workhorse of relational program calculation — B&dM 4.19 and 4.20 — and it is what makes point-free derivations possible at all: it is how a function gets out of the way so the relation underneath can be reasoned about.

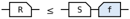
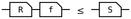
`AOP/A4_2.lean` has both rules for a bare `Allegory` (`map_shunt_right`, `map_shunt_left`), which is where every downstream derivation calls them. They are now also in the diagrammatic layer (`shunt_right`, `shunt_left` in `diag/CB_Derived.lean`) — same statements, so that there is a Lean declaration in `diag` terms for the exporter to draw. Nothing new is assumed: both are proved from `Map` alone.

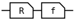
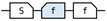
Each rule is an ↔, so each is two chains. The statement of a branch is its header: the hypothesis it is given and the goal it reaches, both drawn from the same Lean ↔ the chain proves.

The proofs, drawn. Each ↔ is two chains, and `diag-export --proof` now splits an `Iff.intro` into its two branches rather than drawing the longer one and dropping the other. The point to watch is that each direction spends exactly one half of `Map` — never both.

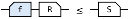
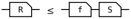
shunting right, ⟹ given  show 

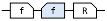
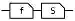
term	picture	the rule that reaches it
`R`		**the start:** `R`.
`≤ R ; (f ; f°)`	≤ 	**`Total f`.** `𝟙 ≤ f f°`, so `f f°` may be inserted after `R`. This is the only thing the forward direction costs.
`≤ S ; f°`	≤ 	Re-bracket to `(R f) f°` and apply the hypothesis `R f ≤ S`. `R ≤ S f°`, which is the goal.

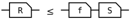
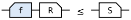
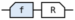
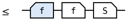
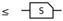
shunting right, ⟸ given  show 

term	picture	the rule that reaches it
`R ; f`		**the start:** `R f`.
`≤ S ; (f° ; f)`	≤ 	Apply the hypothesis `R ≤ S f°` on the left. The two `f`s are now adjacent.
`≤ S`	≤	**`SingleValued f`.** `f° f ≤ 𝟙`, so the pair cancels and `S` is left. The only thing the backward direction costs.

Shunting left is the mirror, and its two chains come out the same shape — `Total f` on the way out, `SingleValued f` on the way back — with the map on the other side throughout.

shunting left, ⟹ given  show 

term	picture	the rule that reaches it
`R`		**the start:** `R`.
`≤ f ; (f° ; R)`	≤ 	**`Total f`.** `f f°` inserted before `R`.
`≤ f ; S`	≤ 	Re-bracket to `f (f° R)` and apply the hypothesis. `R ≤ f S`.

shunting left, ⟸ given  show 		
term	**picture**	**the rule that reaches it**
`f° ; R`		**the start:** `f° R`.
`≤ (f° ; f) ; S`		Apply the hypothesis on the right. The two `f`s become adjacent.
`≤ S`		**`SingleValued f`.** The pair cancels, leaving `S`.

What the rule really is. An adjunction. `R f ≤ S ⟺ R ≤ S f°` says exactly that `f ⊣ f°` in the poset-enriched sense of (35) — `Total f` is its unit and `SingleValued f` its counit — so "`f` is a map" and "`f` has a right adjoint, namely `f°`" are the same statement. That is Lemma 4.8 (`Freyd.Diag.lemma_4_8`), which turns it round: any right adjoint of a map **must** be its converse, by the usual uniqueness argument. The paper's converse half — an arrow with a right adjoint is a map — is the diagram chase that needs Lemma 4.4, and is not proved here for the reason given above.

Where it is spent. In this repo, `map_shunt_right`/`map_shunt_left` carry the relational-product universal property (`AOP/A5_2.lean:177`), division (`AOP/A4_4.lean:370`), the lax naturality conditions (`AOP/A5_7.lean:67`) and the catamorphism work (`AOP/A6_5.lean:372`). It is the single most-used lemma of the AOP layer.

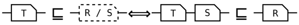
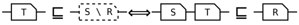
13. Division and negation — the cheatsheet's §8, drawn

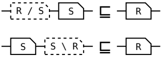
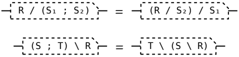
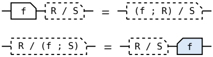
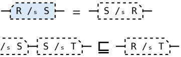
`AOP/Cheatsheet.typ`'s eighth section is B&dM §4.4–4.5: the two divisions, the symmetric division, negation, and Schröder's rule. Every law of it that is about arrows rather than elements is a theorem here — division in `Freyd/S2_30.lean` (Freyd's §2.31), implication, the lexical order and the division map-laws in `AOP/A4_4.lean`, negation and Schröder in `AOP/A4_5.lean` — so every one of them draws. Not one of them is a theorem of the diagrammatic tower, and the pictures say so.

What the notation says here. Dashed means "not a composite of the generators", and in this section everything is dashed: `/`, `\`, `/`$_s$, `⇨` and `~` are what a **division** or a **Boolean** allegory adds to `Allegory`, and no allegory builds any of them. Two shapes carry the whole visible content: the meet `Δ ; (R ⊗ S) ; ∇`, and the tape. So De Morgan comes out as "one box equals a meet of two boxes", excluded middle as "a tape of `R` and `~R` is `⊤`", and Schröder as a composite on one side against a mirrored box on the other — where the converse falls is the one thing to look at.

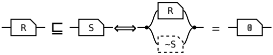
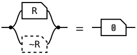
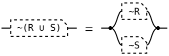
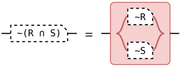
The same caveat as for `°`. A dashed box takes a LABEL, so whatever sits under it is set as text and not drawn: `~(R ∪ S)` is one box, not a tape inside a frame, while the `~R ∩ ~S` opposite it is a real meet. That is why the two sides of De Morgan look like different kinds of picture.

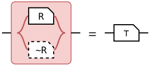
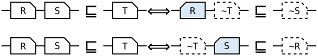
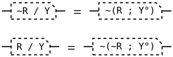
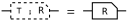
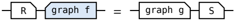
The residual drawn earlier is this `/`. `«≤_residual_iff»` (`diag/FO.lean:171`) and `le_div_iff` (`Freyd/S2_30.lean:54`) are the same Galois statement, `T S ⊑ R ⟺ T ⊑ R/S`, so in a model carrying both structures the two operations agree by uniqueness of adjoints, with neither definition being unfolded. The connection stops there: negation has no counterpart anywhere in the tower, which is why this section leaves the diagrammatic layer and draws the allegory itself.

statement	**picture, from the Lean statement**
Division is the right adjoint of composition: `T ⊑ R/S ⟺ T S ⊑ R` — `Freyd.Alg.le_div_iff` (`Freyd/S2_30.lean:54`). The cheatsheet's `\` reversed, since Freyd divides on the right. Everything else in the division half of §8 is this one line applied twice and read backwards; `div` itself is a field of `DivisionAllegory` with no definition to unfold.	
`T ⊑ S\R ⟺ S T ⊑ R` — `Freyd.Alg.le_leftDiv_iff` (`:309`), the mirror. `S\R ≜ (R°/S°)°` (`:301`), so left division is right division conjugated by the converse and is not a second primitive.	

cancel: `(R/S) S ⊑ R` — the field `DivisionAllegory.div_comp_le` (`:35`); and `S (S\R) ⊑ R` — `Freyd.Alg.leftDiv_comp_le` (`:317`). The counit of each adjunction: the quotient composed back with the divisor lands under the numerator, never on it.	
associate: `R/(S₁ S₂) = (R/S₂)/S₁` — `Freyd.Alg.div_comp_assoc` (`:141`); and `(S T)\R = T\(S\R)` — `Freyd.Alg.leftDiv_comp` (`:332`). Dividing by a composite one factor at a time. The composition is drawn on the left of each equation and swallowed into a label on the right, which is exactly what “the calculus cannot see inside `/`” looks like.	
maps: `f (R/S) = (f R)/S` — `Freyd.Alg.map_comp_div` (`AOP/A4_4.lean:366`); and `R/(f S) = (R/S) f°` — `Freyd.Alg.div_comp_recip_map` (`:376`). Both are the shunting rule of the previous section, spent twice inside an indirect-equality argument. The second is the one that pays: a map moves out of a denominator and reappears as `f°` outside the box.	
symmetric division: `(R/ₛS)° = S/ₛR` — `Freyd.Alg.symmDiv_recip` (`Freyd/S2_30.lean:237`); and `(R/ₛS)(S/ₛT) ⊑ R/ₛT` — `Freyd.Alg.symmDiv_comp` (`:254`). `R/ₛS ≜ (R/S) ∩ (S/R)°` (`:203`) — the cheatsheet’s row. The meet is inside the definition, not inside the statement, so it stays under the label: what the pictures show is that `/ₛ` is converse-symmetric and transitive, i.e. behaves like an equality of columns.	

Negation is ⇨ into 0. `~R ≜ R ⇨ 0` (`AOP/A4_5.lean:44`) on top of `R ⇨ S ≜ ⊔ {X : X ∩ R ⊑ S}` (`AOP/A4_4.lean:100`), so it needs joins of arbitrary families — a **locally complete** distributive allegory — and involutivity `~~R = R` is an extra axiom, `BooleanAllegory` (`AOP/A4_5.lean:123`). The two rows marked *boolean* spend it, and so does the second half of the division-through-negation row.

statement	picture, from the Lean statement
`R ⊑ S ⟺ R ∩ ~S = 0` — `Freyd.Alg.le_iff_inter_neg_zero` (`AOP/A4_5.lean:144`). The bridge every Schröder proof crosses: containment turned into a disjointness, where `comp_inter_zero_iff` can rotate a composite through a converse.	
`R ∩ ~R = 0` — `Freyd.Alg.inter_neg_zero` (`:53`). Non-contradiction, and the only law of the section that holds with no completeness or Boolean assumption beyond the ⇨ that defines ~.	
De Morgan: `~(R ∪ S) = ~R ∩ ~S` — `Freyd.Alg.neg_union` (`:58`). The union is under the ~ on the left, so it prints; the meet on the right is drawn. One equation, both conventions.	
De Morgan, other side (boolean): `~(R ∩ S) = ~R ∪ ~S` — `Freyd.Alg.neg_inter` (`:131`). Needs `~~R = R`: it is `neg_union` applied to ~R, ~S and then un-negated.	

excluded middle (boolean): `R ∪ ~R = ⊤` — `Freyd.Alg.union_neg_eq_top` (:139). `boolean_iff` (:101) is the converse: ~~ is the identity everywhere exactly when this holds everywhere. The two are one axiom, stated two ways.	
Schröder: `R S ⊑ T ⟺ R° ~T ⊑ ~S` — `Freyd.Alg.schroeder_left` (:181); and `R S ⊑ T ⟺ ~T S° ⊑ ~R` — `Freyd.Alg.schroeder_right` (:188). The workhorse of §4.5, and the picture is the clearest in this section: a two-box composite on the left, and on the right the **same** two boxes with one of them mirrored and both survivors dashed. Which factor keeps its orientation is which form you are in.	
division through negation: `(~R)/Y = ~(R Y°)` — `Freyd.Alg.neg_div` (:275), which holds without the Boolean axiom; and `R/Y = ~((~R) Y°)` — `Freyd.Alg.div_eq_neg_comp` (:293), which needs it. The cheatsheet's “neg. division” row. In a Boolean allegory division is not independent structure at all: the second equation defines / from ~, ° and composition.	
lexical order: `⊤ ; R = R` — `Freyd.Alg.topHom_thenRel` (`AOP/A4_4.lean:304`). `R ; S ≜ R ∩ (R° ⇒ S)` (:269) — compare by R, break ties by S. Preorders are closed under it (`thenRel_transitive`, :282) and ⊤ is its unit, which is the cheatsheet's “unit Π”.	
Galois, point-free: `f(x) ≼ y ⟺ x ⊴ g(y)` is `R f° = g S` — `Freyd.Alg.RelSet.galoisPF` (`AOP/A4_7_Galois.lean:36`), with `galois_iff` (:63) proving it equivalent to the pointwise form. Stated in `Rel(Set)` rather than in an allegory, because the pointwise side needs elements; `graph f` is f as a relation. The picture is the point-free form's whole advantage — two composites, no quantifiers.	

The measurement. Twenty-one pictures, and a generator appears in five of them: a meet in `le_iff_inter_neg_zero`, `inter_neg_zero` and `neg_union`, a tape in `neg_inter` and `union_neg_eq_top`. The other sixteen are boxes threaded on a wire. That is not a defect of the rendering — it is the content: ∩, ∪, ° and composition are what the Frobenius generators build, and §4.4–4.5 is made of the four operations that they do not.

What is not drawn, and why. The cheatsheet's pointwise row, `x (T\U) y ⟺ ∀w. w T x ⟹ w U y`, quantifies over elements and has no arrow to draw; the strict part `|R| ≜ R ∩ ~R°` has no definition in the repo to export. Both are one-line additions if a later section wants them — the first as a `Rel(Set)` lemma beside `galoisPF`, the second beside `neg`.